

Shaft Sinking Special Method

15/05/2020

Specialized Method of Shaft I

- *Specialized method of shaft sinking are used when following conditions are encountered.
- *Incompetent, water bearing rock (quick sand)
- *Weak unstable, soil type rocks.
- *Competent fractured rocks, with high water inflows above $0.5\text{m}^3/\text{min}$.
- *There are numerous special methods are available to suit conditions.

Wooden Pile Wall I

*Wooden piles are used for protecting shaft in loose & unstable ground.

*This method can be applied in circular shaft, rectangular shaft.

$$D_g = D + 2(d + 2f + c + f)$$

D_g > initial diameter of shaft collar

D > shaft diameter

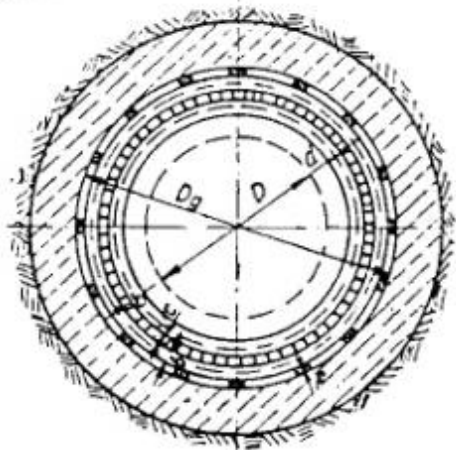
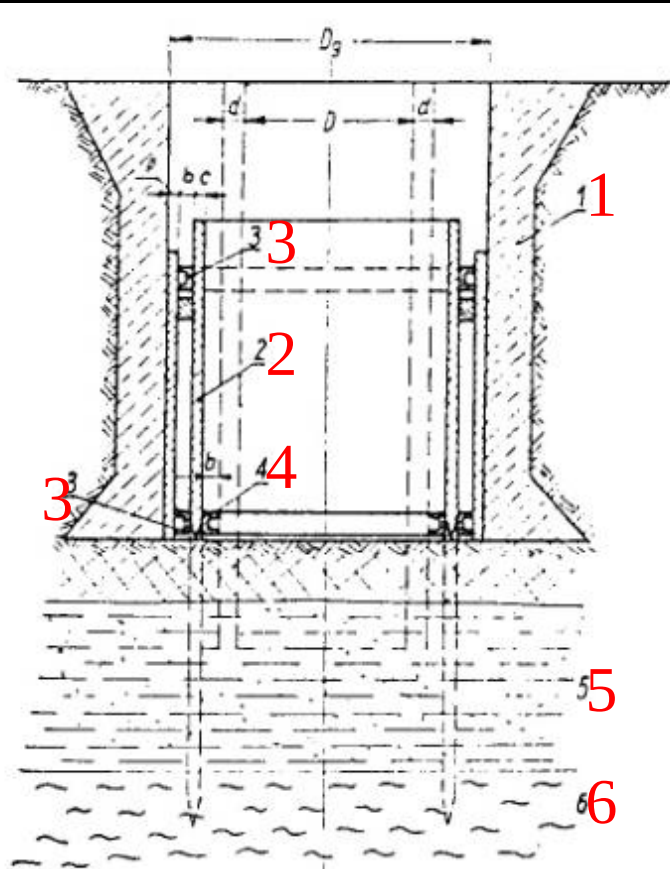
d > permanent lining thickness

b > width of guide ring

c > thickness of driving poles

f > thickness of guide beam

Vertical Wooden Pile Wall



1- Shaft Collar

2- Piles

3- Outer guidance ring

4- Inner guidance ring

5- Water bearing strata

6- Under clay strata.

D_g - Initial diameter of shaft collar

d - Permanent lining thickness

D - Final shaft diameter.

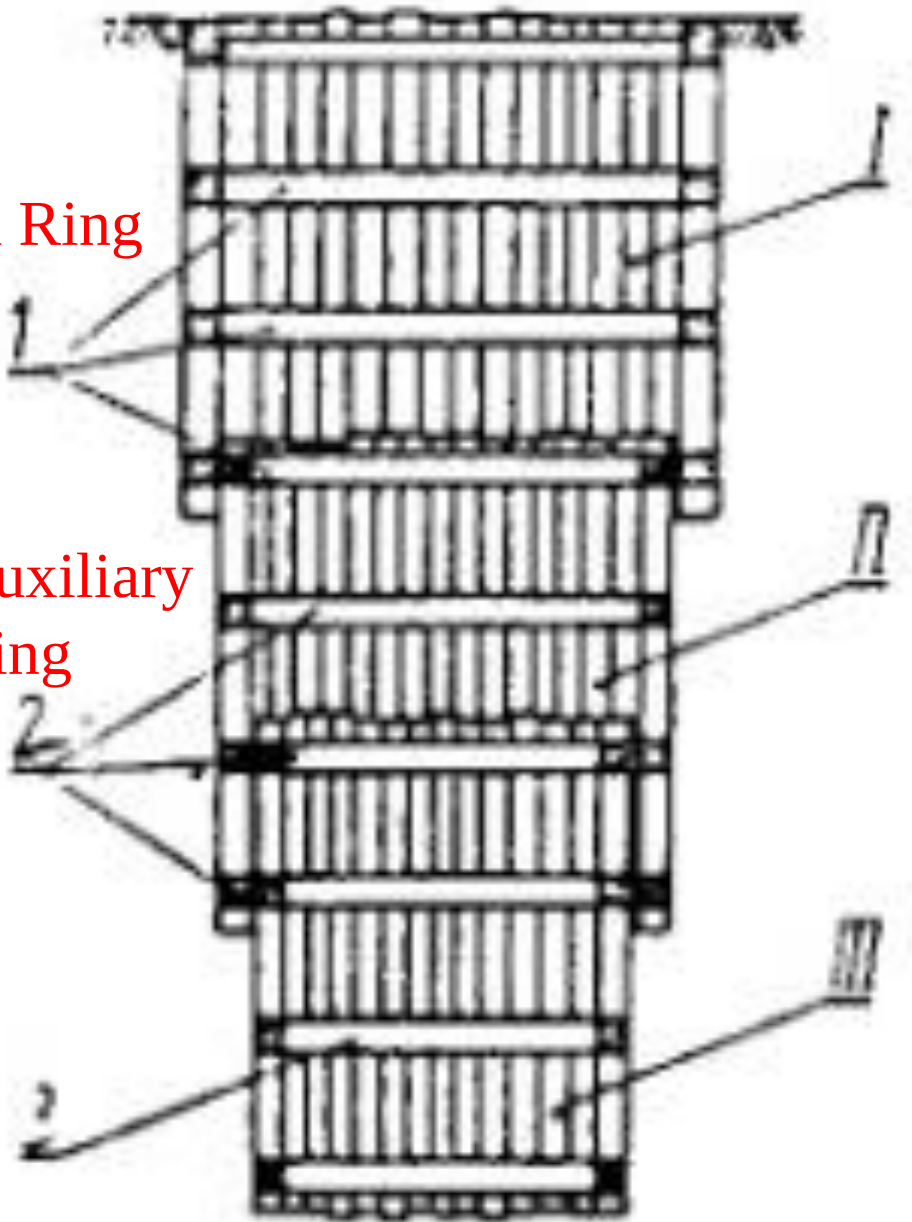
f - Thickness of guide beam.

Wooden Piles Rectangular Shaft

Main Ring

Auxiliary
Ring

Wooden piles wall
for rectangular
shaft vertical
method with
diminishing
lateral section
I,II,III,IV
subsequent
section.



Wooden Pile Wall for Rectangular Shaft Oblique method, with steady lateral section

3 Driven Piles

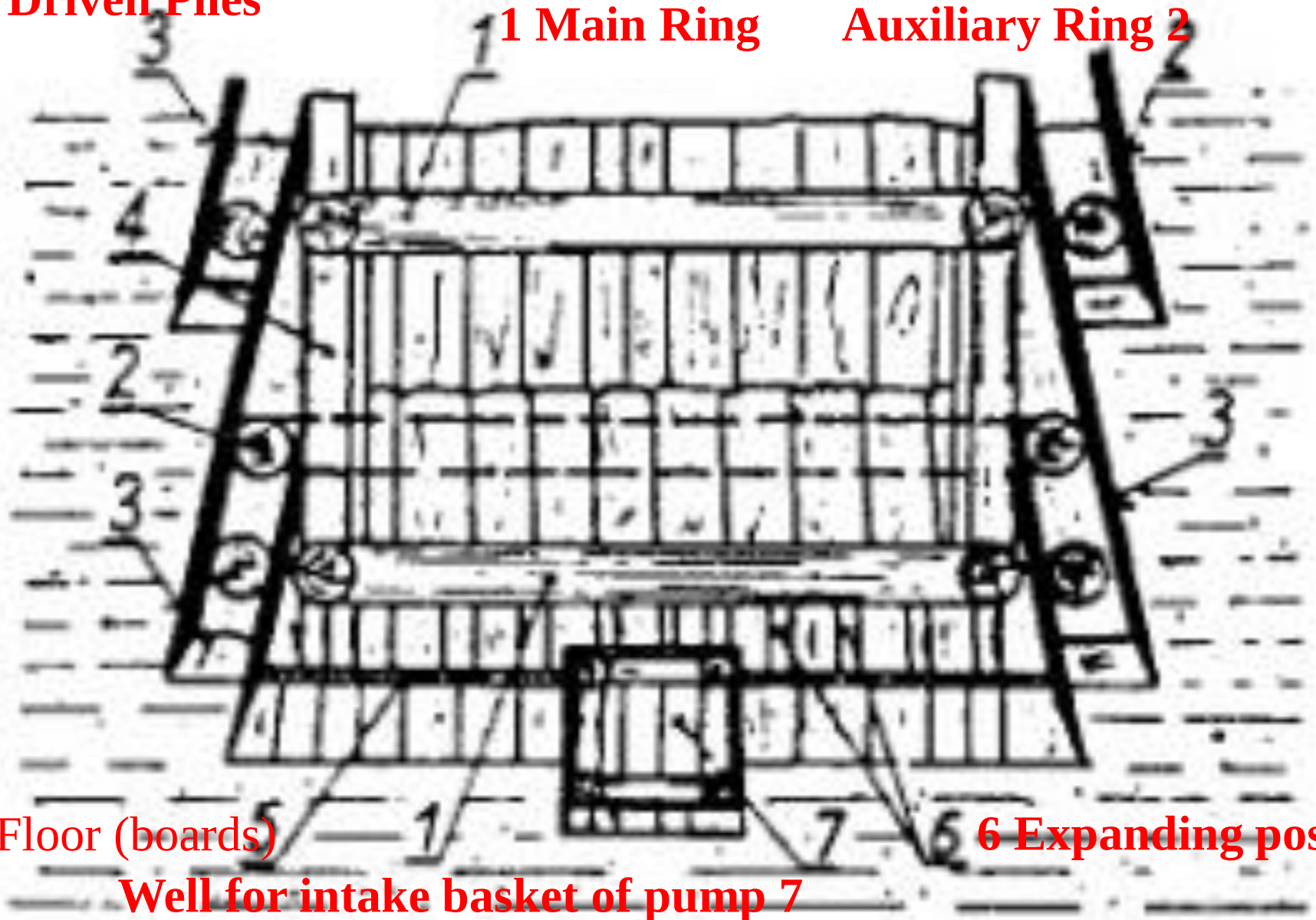
1 Main Ring

Auxiliary Ring 2

5 Floor (boards)

6 Expanding post

Well for intake basket of pump 7



Sheet Pile Wall (Steel) I

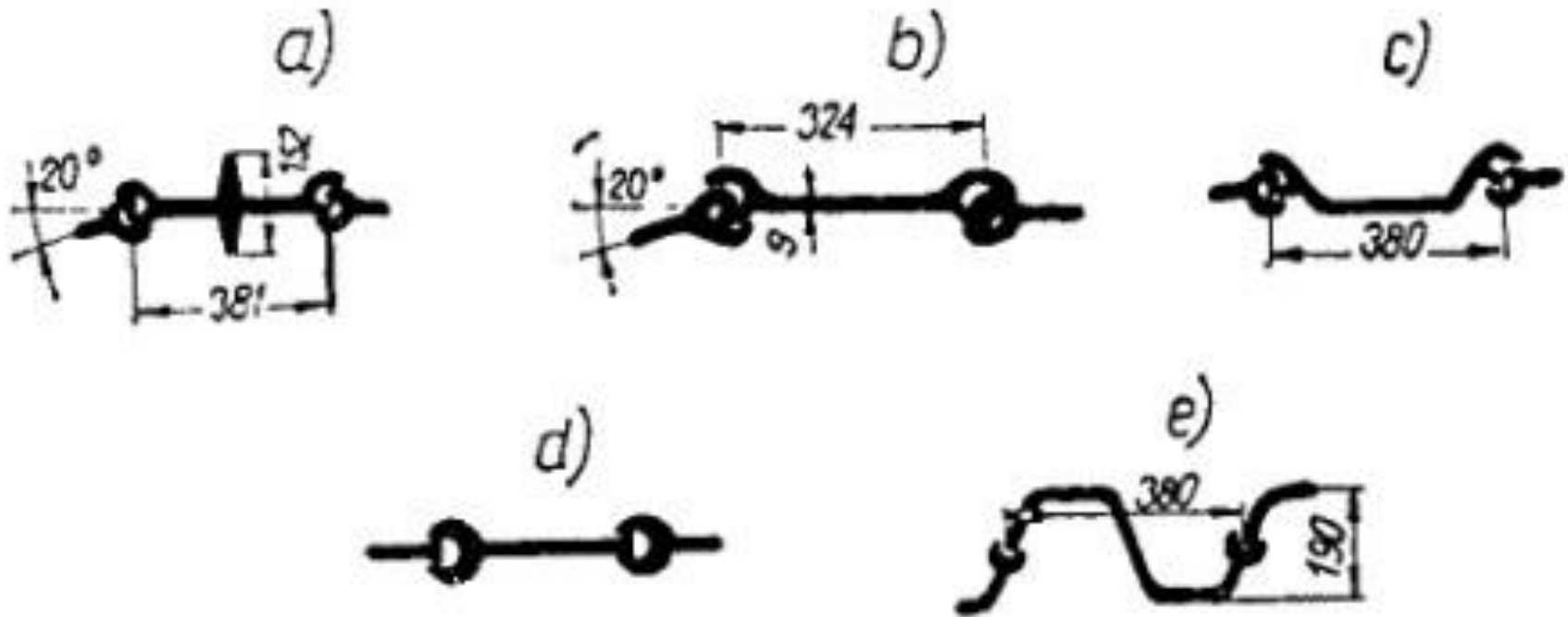


Fig. 17.4.38. Sheet piles.

a to e) different type of joints of sheet pile walls. Lengths up to 15 m. Joints allow consecutive sheet pile to deviate 20°. Dimensions in cm. Conversion factors: 1 in. = 2.54 cm, 1 ft = 0.3048 m.

Concrete Pile Wall I

Hammered down pile

- *Concrete pile in ground by pile driver.
- *Piles are driven beside each other cutting into side of neighboring concrete pile.
- *This create a tight wall, preventing any loose rock penetrating excavation.

Concrete Pile Wall II

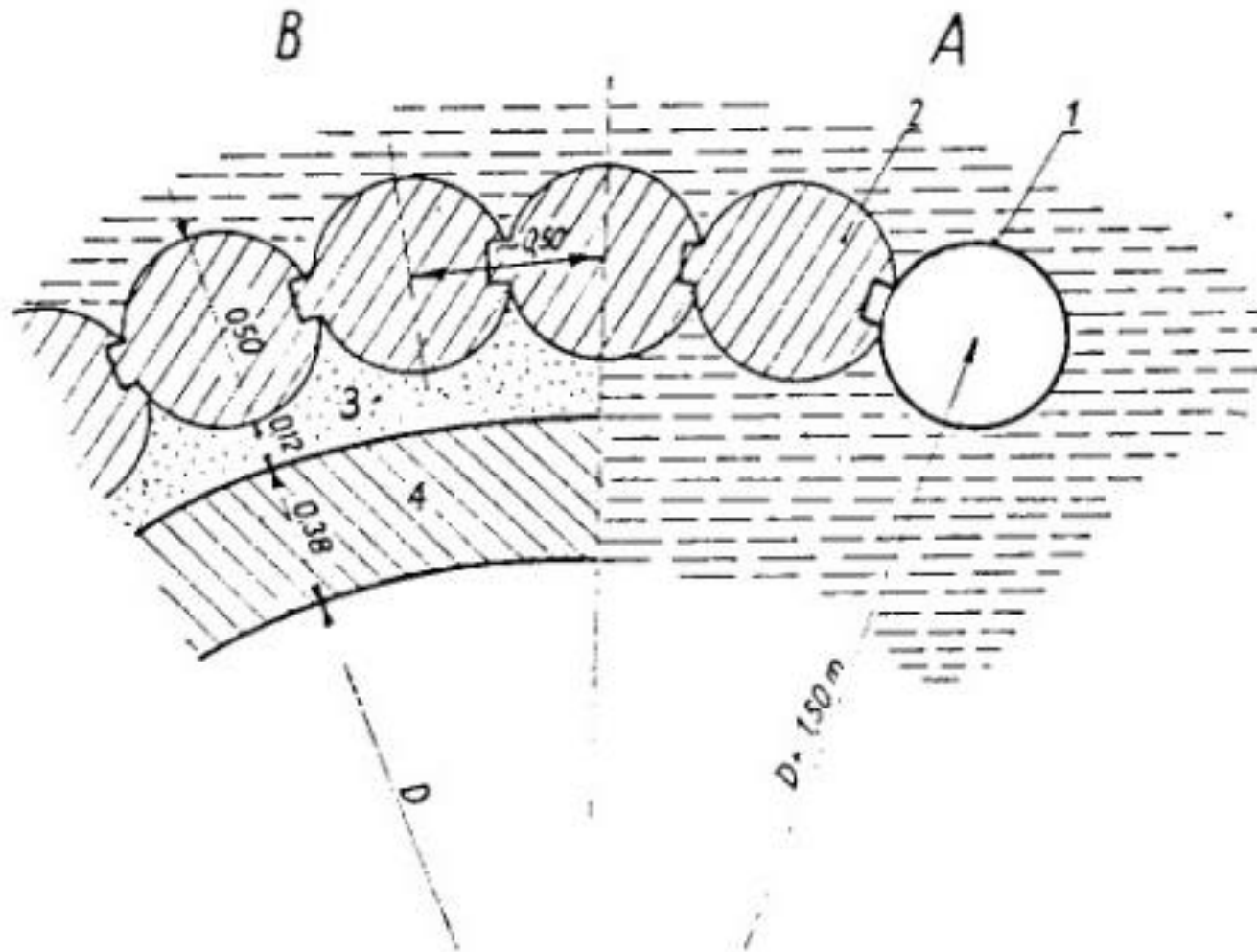
Drilled piles

- *Drilled piles are of 0.5 to 1.2m in diameter.
- *Outer steel pipe is placed in to ground with help of drilling rig.
- *Where ground does not assure stability of side wall, polymer slurry is used to prevent failure of hole.
- *Concrete used for pile is low water cement ratio (0.4).
- *To assure tightness of concrete, it is placed in batches & compacted with hammer.
- *For longer pile, reinforcement is used.

Concrete Pile Wall III

*Limit of piling depth is 40 m.

Concrete Pile Wall I



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Fig. 17.4.39. Lateral section.

a) driving construction

b) with shaft lining erected with protective pile wall.

Conversion factor: 1 ft = 0.3048 m.

Concrete Pile Wall I

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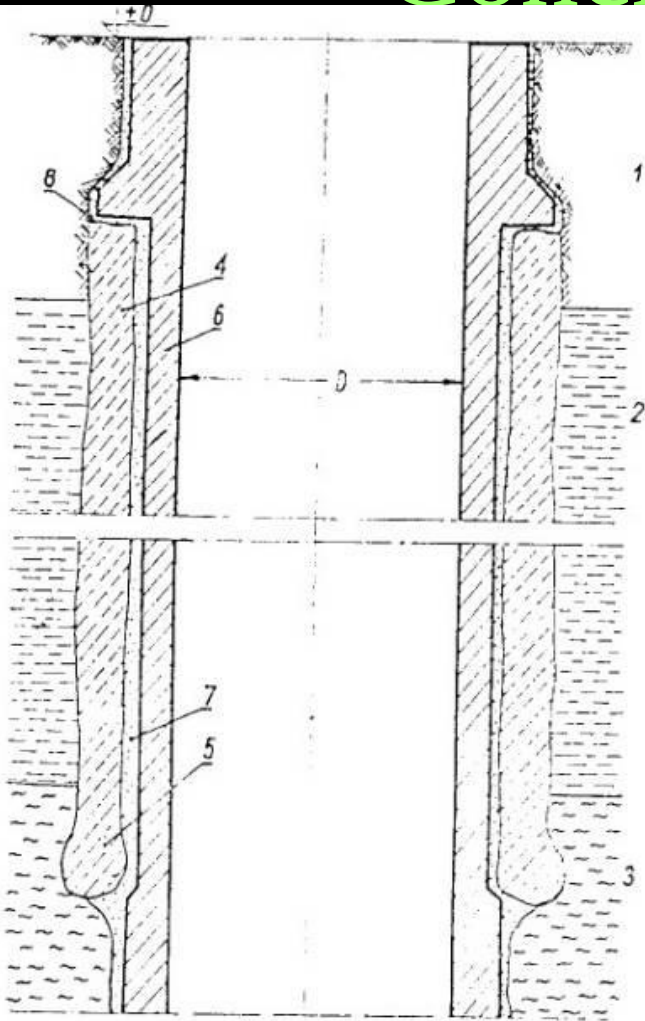


Fig. 17.4.40. Vertical section through the shaft.

1. dry ground
2. water-bearing ground
3. nonpermeable clay strata
4. pile wall
5. base of the pile
6. shaft lining
7. inner layer of concrete
8. upper end of the pile wall.

Caisson Method I

- *Cassion method is used in shaft sinking, when surrounding rock / soil is very watery and unstable.
- *In cession method, shaft sinking is done with help of heavy steel shutter.
- *Steel shutter moves down protecting weak side wall, as material is removed from bottom of shaft.

Caisson Method II

- *Caisson moves down by its own weight, some time additional weight is added, if required.
- *Caisson can also be pushed down by Jack.
- *For caisson to move down, it is required that downward force should exceed frictional force of side wall.
- *Some time frictional force is reduced by injecting water in between caisson & soil/rock.
- *It is possible to diminish friction by injecting water at 210 to 320kPa or more than hydrostatic pressure, through holes in cutting toe.

Caisson Method III

- *Cession method to work, following conditions are to be satisfied.
- *When water being pumped $Q > f$
- *With out pumping water $Q > F + G$

Caisson Method III

*Relations

$$Q = \pi(D_{2sq} - D_{1sq}) SH/4 + B$$

$$G = \pi(D_{2sq} - D_{1sq}) H_w S_w / 4 + V_b S_w$$

$$F = \sum_{i=1}^{i=n} f_i = \pi D^2 \sum_{i=1}^{i=n} H_i F_i$$

Caisson Method IV

*

Q = weight of caisson

Q_1 = weight of caisson partly submerged to depth H

F = resistance by friction

D_2 = outer diameter in m

D_1 = inner diameter in m

S = specific gravity of lining material

H = height of caisson in m

H_w = height of caisson submerged in m

B = weight of cutting toe

V = volume of cutting toe

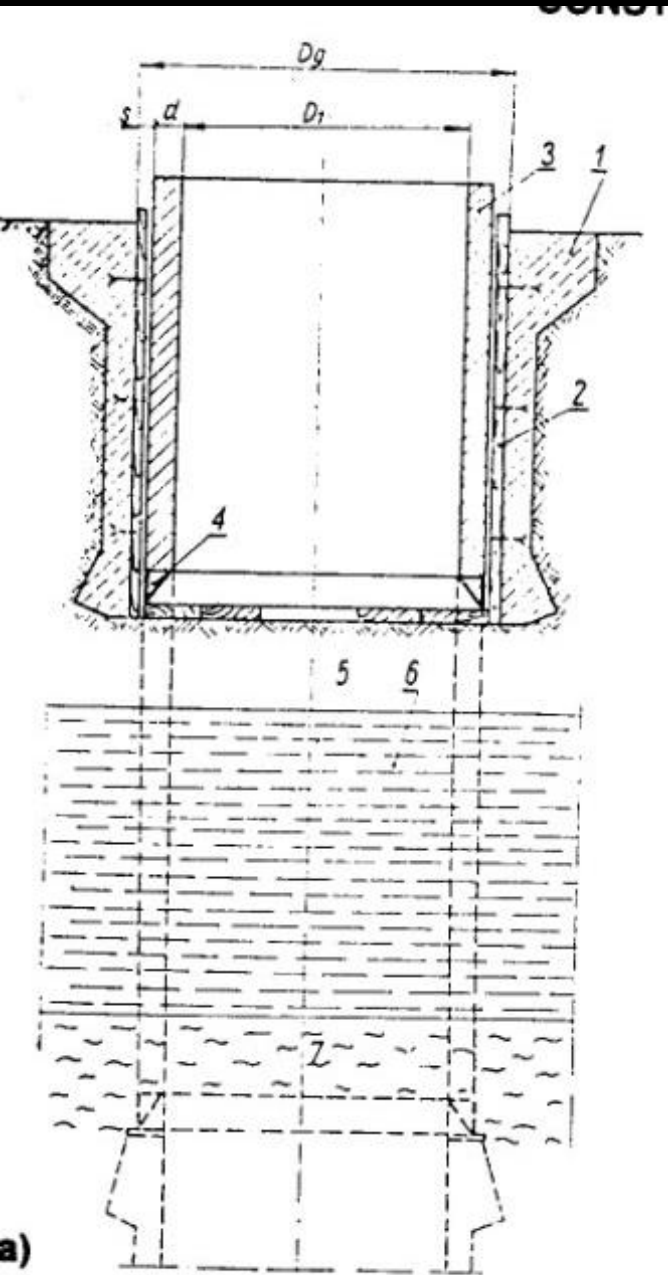
Caisson Method IV

S_w =specific gravity of water t/m^3

F_i = frictional force per unit in a particular strata in t/m^3

H_i =thickness of particular strata in m

Shaft Collar For Cession Method



1- shaft collar

2- guide beam

3- brick lining

4- cutting toe (boot)

5- dry strata

6- water bearing strata

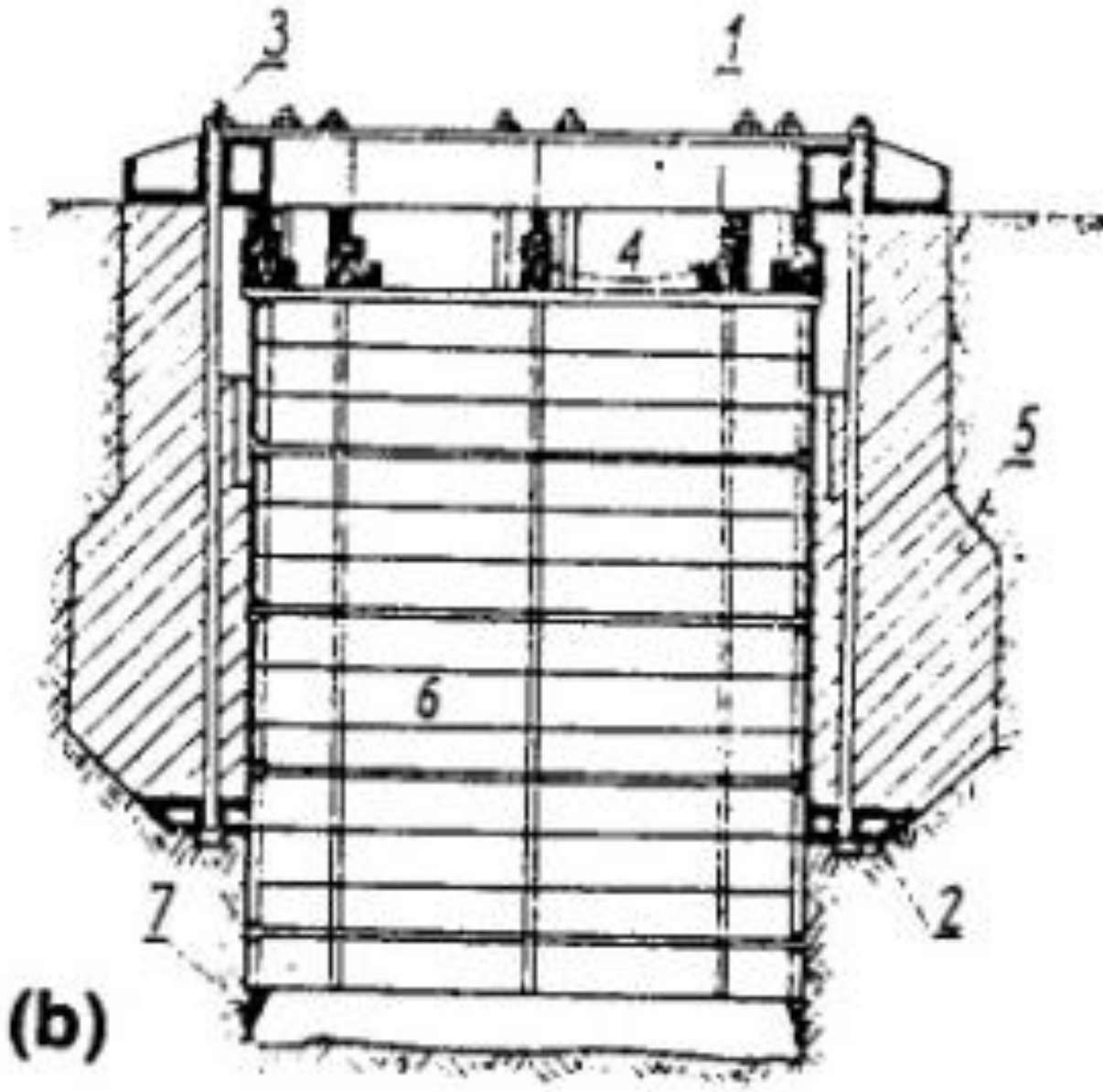
7- non permeable (clay) strata

D_g - initial diameter of shaft collar

D_1 – final diameter of shaft

d - thickness of cession

Shaft Collar For Jacking at cast iron tubing ring



1- reaction ring for jacking at cast iron tubing lining.

2- auxiliary reaction ring

3- pull rod connecting rings with wall of collar

4- hydraulic jacks

5- shaft collar wall

6- water bearing strata

7- non permeable strata

Caisson Method Compressed Air I

- *Range of application of this method is 30 meter, because maximum air pressure should not exceed 350kpa.
- *Decompression chamber is required to avoided nitrogen buildup in blood for face workers.
- *Compressed air in caisson forces water back in to ground.
- *Face remains dry to work with.
- *Air pressure should be adjusted such that water will stay about 100 to 400 mm.

Caisson Method Compressed Air II

- *In this method, lock chamber or chambers are required to transfer of crew & material handling.
- *Thickness of roof at working chamber is calculated according to maximum pressure with safety factor 1.5.
- *Lock dock normally has one chamber to accommodate hoisting facilities, personal traffic & material hoisting.

Caisson Method Compressed Air III

- *To assure proper sinking, vertical force downwards must be larger than combined effect of friction & vertical force acting up wards caused by positive air pressure in caisson.
- *Achieved rate of shaft sinking by this method is 20m /month maximum 45m in a month in coal mine.

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Depression of Water Table I

- *Determination of number, spacing, diameter & depth of wells.
- *Calculation of radius of depression cone, amount of water to be pumped out & height of remaining wet portion of water bearing strata.
- *Equipment, hardware & pumping facilities, choice of auxiliary method to go through wet reaming of water bearing strata.
- *Pumping water from 3 to 6 depression wells drilled around planned shaft perimeter.

Depression of Water Table II

- *Wells are placed on circle about 4m larger than diameter of shaft excavation.
- *Water is drained to existing opening in mine below through drill holes.
- *Water can also be drained to underground water absorbing strata (dry sand under water bearing strata).
- *Wells are drilled with 300 to 500mm diameter, without drilling mud.
- *Wells are placed uniformly around future excavation.

Depression of Water Table III

- *Within water bearing strata, well are equipped with filters perforated by round or elongated holes 5 to 10mm in diameter, spaced in checker design, with an out side screen to filter sand particle.
- *Construction of filter must be suitable for characteristic of strata to be drained.
- * Submersible drainage pump have a wide range of capacity up to 350m³/hour.
- *Calculation of well efficiency are based principle of flow through porous media.

Depression of Water Table V

- *Design is to determine a pumping regime to achieve a dry out zone surrounding shaft.
- *This is created by cumulative effect of depression cone around particular drainage well.
- *Movement of water through a porous media is given by Darcy formula.
- *Co-efficient of filtration k may be expressed in following unit m/s, cm/s, m/day
- *Co-efficient of filtration for particular rock depends on rock structure, grain size etc.

Depression of Water Table VI

- *Best drainage results are obtained in coarse & mid size sand with $k=10-100/\text{day}$.
- *Drainage method can be successfully combined with other method (compressed air caisson method).
- *It allows caisson method to work with low air pressure in working chamber.
- *With fine sand, there is possibility of sand getting washed away with pumping.

Depression of Water Table VII

- *This can cause subsidence.
- *Adequate filters are to be applied to control silting problem.

Depression of Water Table IV

$$V = k \cdot dh/dl = k_i + k_h/l$$

V > mean velocity of flow over unit time through a unit area or mean filtration velocity in m/sec.

$H = h_2 - h_1$, difference of water level in ground (depression) created by pumping water or head loss during flow.

l > distance in which head loss appears in m.

k > co-efficient of filtration (permeability factor) in m/sec

$i = h/l = (h_2 - h_1)/l$ or head loss over unit distance (hydraulic gradient).

Shaft Sinking With Grouting I

- *Before starting grouting, following options have to be finalized & information's to be collected.
- *Whether grouting will be done from surface or under ground.
- *Determination of grouting depth.
- *Determination of length of grouting section.
- *Number of grouting holes & their respective distances.
- *Diameter, depth & angle of dip & applicable drilling method.
- *Contraction & calculation of grouting plug.

Shaft Sinking With Grouting II

- *Grout type to be used, its density & admixtures.
- *Injecting system & required equipment.
- *Determination of working pressure to be applied in particular section & approximate calculation of grout consumption.
- *Number of controlled holes & their locations.
- *Set of drawing corresponding to different issues.
- *Number of characteristics of water horizon.
- *Chemical analysis of water flow & determination of their degree of corrosiveness.
- *Flow velocity.

Shaft Sinking With Grouting III

*Coefficient of filtration.

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Shaft Sinking, Grouting System I

- *Properly performed grouting seals rock mass in vicinity of shaft.
- *Grouting makes sinking by conventional method possible.
- *Grouting is performed from surface, when water bearing strata are relatively thick & close to surface.
- *Grouting is done from shaft face, when strata to be grouted are located at such a depth that is not feasible to drill long holes from surface.
- *Grouting is most successful in water bearing, competent but fractured rocks.

Shaft Sinking, Grouting System II

- *Grout fills fractures & cracks, creating a sealed zone around shaft to be sunk & protecting against water flow.
- *Grouting for sealing is often applied to improve tightness of lining in existing shaft & to eliminating voids behind linings, as well as to seal rocks in immediate vicinity of linings.

Shaft Sinking, Grouting From Surface I

- *Grouting holes with an initial diameter of 75mm to 150mm are drilled vertically around fractured shaft on a circle with radius of 3-4m larger than excavation diameter.
- *Hole size should be as small as possible to minimize drilling cost & to maintain sufficient velocity of grout flow to avoid sedimentation of cement particle.
- *Distance between holes usually in range of 2 to 4m.
- *Holes are without casings, excluding part in loose rocks (usually near surface).

Shaft Sinking, Grouting From Surface II

- *Entry to hole has a central guide pipe 3-5m long.
- *Depending on thickness of zone to be grouted, one or more grouting sections are chosen.
- *Practical length of section is 25-50m.
- *In rock with uniform fractures, distribution & porosity, longer sections are used.
- *About 24 hours after upper section is grouted, hole is redrilled, in relatively soft ground.
- *Hole is extended by length of next section & then grouted after drilling holes are flushed with water until clear water comes out from hole.

Shaft Sinking, Grouting From Surface III

*Holes should be straight & with greater depth; any deviation should be recorded.

Shaft With Grouting from Surface I

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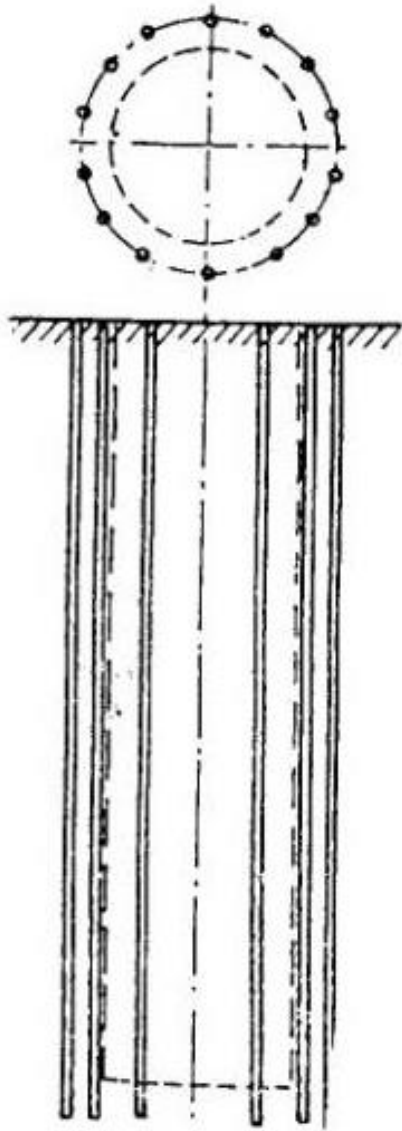


Fig. 17.4.42. Schematic of distribution of grouting holes drilled from the surface.

Shaft Sinking, Grouting From Shaft I

- *Grouting from shaft face is recommended when water bearing strata occurs below 100m & their thickness does not exceed 70m.
- *Length of section varies between 12 to 25m.
- *Holes are drilled along length of grouting section.
- *Drilling is done with help of heavy drill hammer or small rig, depending up on space available.
- *Holes are directed at an angle in vertical plane to get hole bottom 1-2m away from future excavation.
- *Holes are tilted in a tangential direction to increase probability of encountering fracture.

Shaft Sinking, Grouting From Shaft II

- *Distance between hole entries are 0.8 to 1.5m.
- *Spacing between circles of holes is 1.5 to 2 to 3m.
- *Number of holes drilled depends on rock character, shaft diameter and hydro geologic conditions.
- *Number of hole varies between 10-20-30.
- *Initial drilling size is 50-70mm, end size is 30 to 50mm.
- *First one-half of total number of holes are drilled & grouted, during hardening of cement, second half is drilled & grouted.

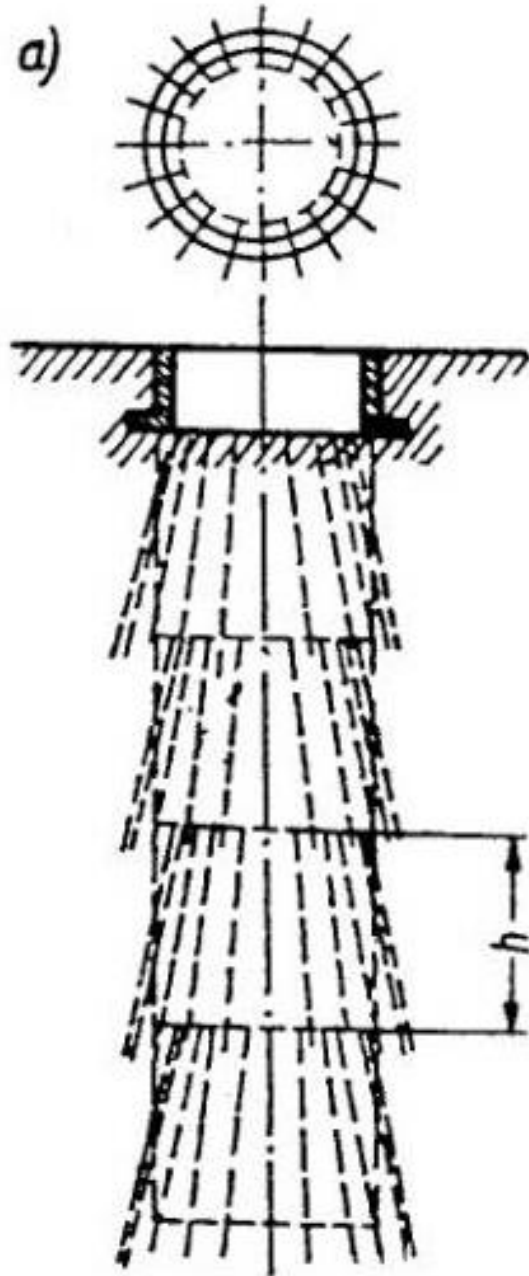
Shaft Sinking, Grouting From Shaft III

- *All holes are drilled & grouted one after another, starting with hole with larger water in flow.
- *A pair of holes lying on opposite side are drilled & grouted.
- *Pair lying in plane perpendicular to first one is processed.
- *This system is most often used, being recognized as best.
- *Initial maximum distance between holes, diminishing probability of grout penetration from one hole to another hole.

Shaft Sinking, Grouting From Shaft IV

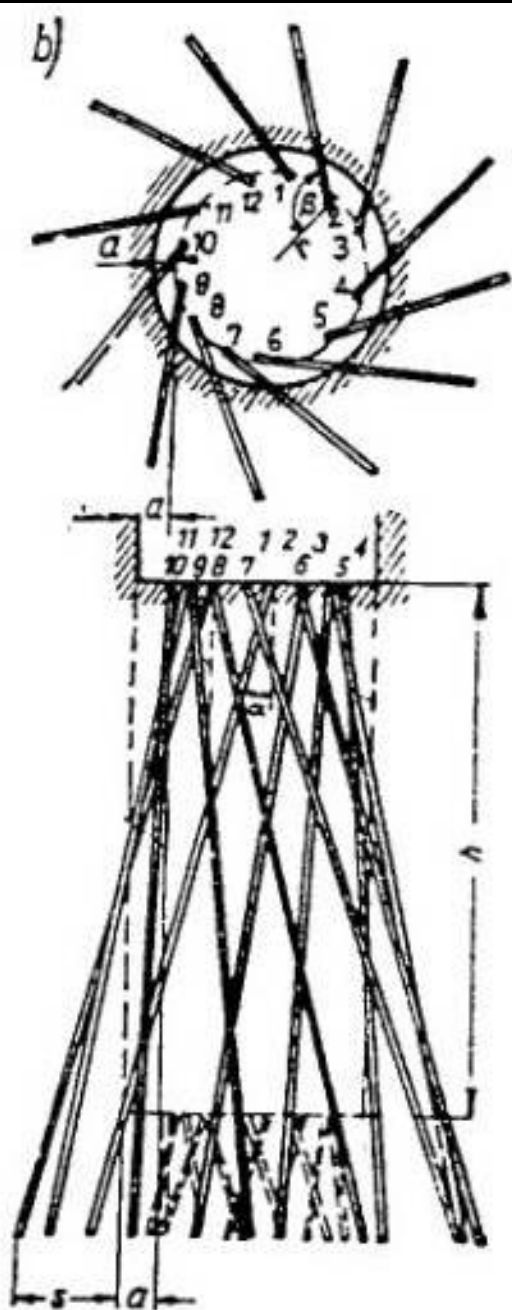
*Entrance of grout holes are equipped with entrance pipe with diameter 50 to 75 mm & length of 1.5 to 2m.

Shaft With Grouting From Shaft I



- *Grout hole distribution drilled from shaft face.
- *Holes are projected away from face.
- *Rounds of holes are drilled at 12 to 25m interval (h)

Shaft Complementary Grouting I



- *Grout hole distribution drilled from shaft face.
- *Holes are projected diagonally away from face.
- *Rounds of holes are drilled at 12 to 25m interval (h)
- *Distance in between two holes is 1 to 1.5m (a)
- *Holes are spaced about 2 to 3 m from shaft wall.

Shaft Sinking, Grouting Plug I

- *Grouting plug are put in shaft bottom to prevent water & grout inflow in to shaft.
- *Entrance pipes to holes are secured firmly by grouting them in plug.
- *Plug can be either natural or artificial.

Shaft Sinking Natural Plug I

- *Natural plug is composed of portative layer of grouted rock 3 to 5m thick.
- *Thickness depends on strength of rock & expected pressure (water & grout).

Shaft Sinking Artificial Plug I

- *Artificial plug is used more often.
- *Artificial plug is made of concrete or reinforced concrete.
- *It is constructed at shaft bottom, 2-4m from water bearing strata.
- *After completion of grouting, plug is dismantled with explosive.
- *Sinking progresses through cemented section.
- *Sinking is suspended 2 to 4 m above end of grouted zone.

Shaft Sinking Artificial Plug II

- *Shaft lining is constructed & this procedure with plug is repeated until shaft passes whole water bearing zone.
- *During sinking of shaft through grouted zone, test drill holes are drilled to monitor hazardous water inflow.
- *Artificial plug can be flat, spherical or multi stage.
- *Appropriate design calculation are required before starting work.

Shaft Sinking Construction of Plug I

- *Construction method depends on volume of water inflow to shaft.
- *When inflow of water is small, it can be controlled by pumps, a wooden chest with volume of 1m^3 at bottom.
- *Holes of 30mm diameter are left for water to enter chest.
- *In chest, a vertical drainage pipe is installed with a diameter larger than suction of pump.
- *Vertical drainage pipe is 1m longer than thickness of plug.

Shaft Sinking Construction of Plug II

- *On shaft bottom, a layer of gravel 0.6 to 1m thickness is placed for water filtration.
- *Through drainage pipe, suction pipe of pump is lowered so that during plug construction, water inflow is controlled.
- *After plug has been hardened, gravel layer is grouted through pipe previously installed with in plug.
- *Grouting holes are drilled through entrance pipe already installed in upper part of plug (during placing of concrete).

Shaft Sinking Construction of Plug III

- *When water inflow is too great to control with pumps, an under water plug is constructed.
- *Concrete is pumped on shaft bottom through tubing to form layer of 0.3 to 0.4m thick.
- *It is important to keep outlet of rising tubing below concrete surface.

Shaft Grouting Plug

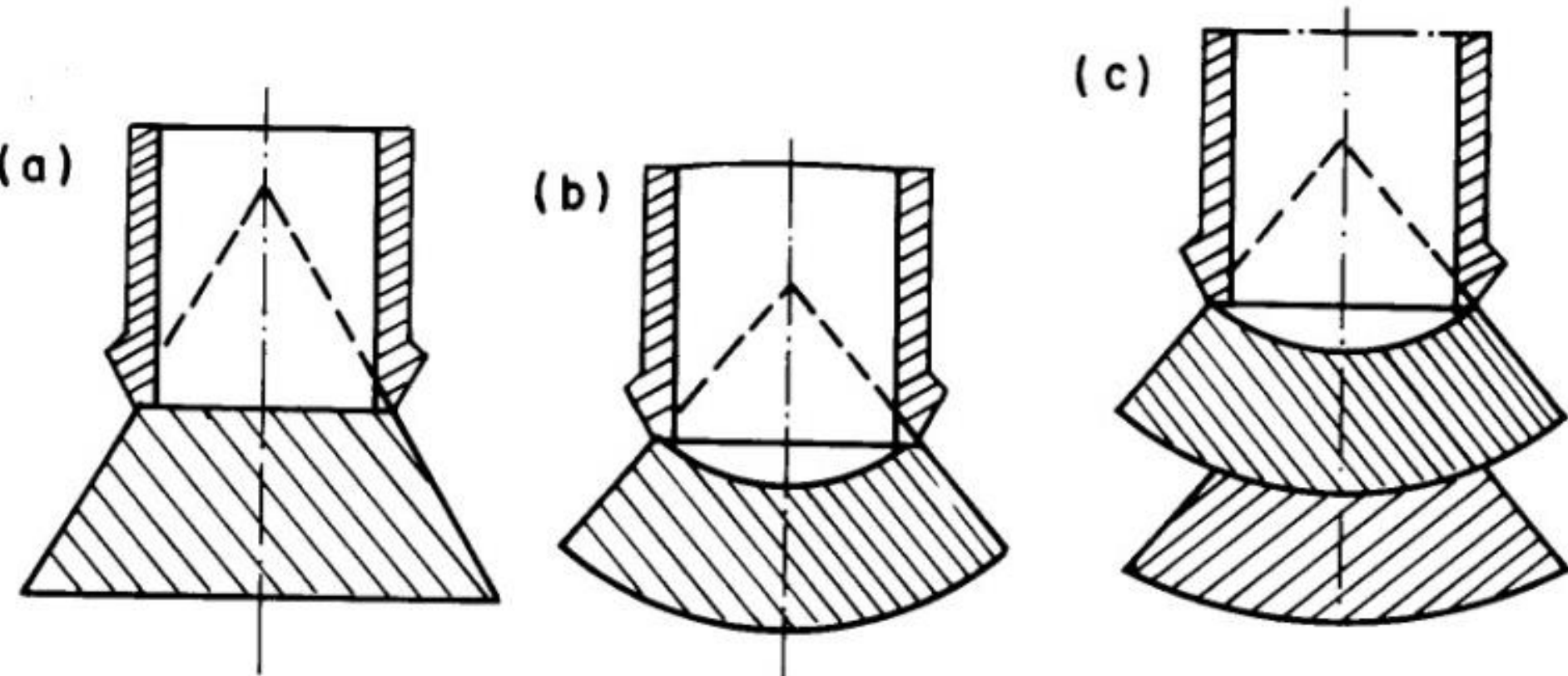
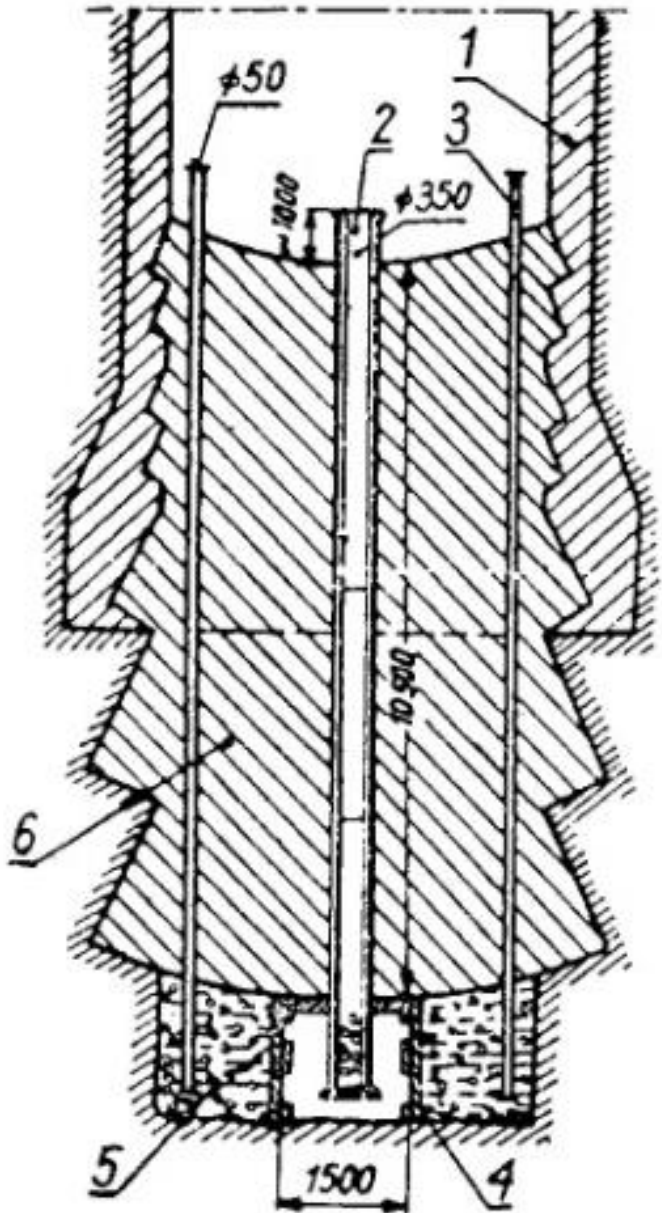


Fig. 17.4.45. Shaft grouting plugs.

- a) flat
- b) spherical, one stage
- c) spherical, double stage.

Shaft Grouting Plug I



Construction method with grouting plug

1- Shaft permanent lining.

2- Drainage pipe 350mm dia

3- Grouting pipe 50mm dia

4- Wooden chest with hole for water collection.

5- Gravel or crushed stone.

6- Double stage concrete plug.

Shaft Sinking Grouting Selection I

- *Selection of cement depends on.
- *Character of water (salty or fresh).
- *Mineralization & chemical corrosiveness.
- *Value of filtration factor.
- *Proper choice is made on basis of results obtained from chemical analysis & pumping test.

Shaft Sinking Density of Grout I

- *Choice of grout density depends on water abnormality measured in hole.
- *It is determined by pumping clear water for 20 minutes period to grouting hole with head exceeding 300 to 500 kpa hydrostatics pressure.
- *Water absorbed in L/min can be determined by

$$q = Q/Hh$$

Q> water volume being pumped.

q> water absorbed in L/min

h> vertical height in m of hole where absorbility is being measured.

H> water head in m

Shaft Sinking Density of Grout II

- *In presence of salty water, same salt concentration in water is used to prepare grout.
- *Grout density depends on character of discontinuities in rock mass & applied grout pressure.
- *This density changes over wide limits ranging from 3 to 50%.
- *When large cavities are encountered then a thick grout (with sand fillers) is first pumped in, followed by a thinner mixture constituting of water, cement only.

Shaft Sinking Density of Grout II

- *For rock with small cracks, thin grout (3 to 5%) cement is preferred.
- *Density of grout must be changed to match absorbability of hole.
- *In rocks while very small fissures were grouted grouting can not be practiced will even under high head, special chemical are used to decrease friction.
- *To increase absorbability, small explosive charges are used in hole, based on need.

Shaft Sinking Density of Grout III

- *For larger fissures, dense grout (25 to 30%) is injected.
- *To decrease absorbability & lessen cost of grouting operation, clay – cement – sand is used with cement – clay – sand in corresponding proportion of 1:1:1.5 or 1:1:2 respectively.
- *By using clay cement grout, cement is spared without scarifying tightness of rock mass.
- *It is advisable before planning a grouting operation to carry out a study & laboratory examination of rock samples to determine an optimum grouting regimen.

Shaft Sinking Density of Grout IV

- *Hardening time of cement should be selected to allow grout penetration & injection to desired radius of penetration.
- *Volume of injected grout is justified that seals rock to meet minimum design requirement.
- *In case of high permeability, a substantial loss of grout may easily occur.
- *This may raise cost of shaft sinking
- *An admixture to speed time of setting up cement are required, especially in presence of water flow.

Shaft Sinking Grout Equipment I

- * Main component of grouting set up are provided below.
- *Cementation pump (two unit) for grout injection.
- *Mechanical mixture, where grout is prepared.
- *Grout Tank.
- *Flexible pressure hoses.
- *Injectors.
- *Pipe, valves, manometer, tools etc.
- *Cementation pumps are able to change head & output wide range.

Shaft Sinking Grout Equipment II

- *Piston or plunger pumps with a capacity of 0.2 to 1m / minutes are commonly used at working pressure up to 15Mpa.
- *Pumps could have been driven by electric or compressed air motor drive.
- *Where grouting operation is carried out at substantial depth, hydrostatic pressure of grout can be sufficient for grouting.
- *In all other conditions, cement grouting pumps are to be used.

Shaft Sinking Grout Pressure I

*Applied grout head should be established separately for each section of cementation.

*When grouting is done from surface, maximum pressure p in Pa is provided by following formula

$$p = Hdk$$

p > Maximum pressure in Pa

H > depth counted from surface to top of strata in m

d > density of rock (1.5 to 2.0 t/m)

k > co-efficient of cohesion of strata above grouting zone (2 to 3)

Shaft Sinking Grout Pressure II

- *When grouting from shaft bottom, required pressure has to be found by taking in to strength of shaft lining.
- *A ground pressure is set at 1 to 3 MPa, higher than hydrostatic head.
- *Calculation is essential to ensure that lining is not threatened by grout pressure.
- *Shaft lining should be kept under observation during cement injection.
- *Observation is done in control pipe with plugs installed in lining (3 to 5m) above shaft bottom.

Shaft Sinking Grout Pressure III

- *Prior to injection, a tightness test of installations is done with pressure 1.5 times maximum scheduled pressure.
- *Grout hole is flushed out with water to clean out silt & clay until return water is clean.
- *Only after these steps are taken to execute grouting operation.
- *When grouting is terminated, cut off valve is closed, grouting installation should be flushed with water, to prevent grout from solidifying in pipe.

Shaft Sinking Grout Pressure IV

- *After 24 hours, a grout hole can be drilled again.
- *In case of water after out flow, grouting procedure is repeated.
- *Often one injection may not be sufficient & one or more additional treatments are required.

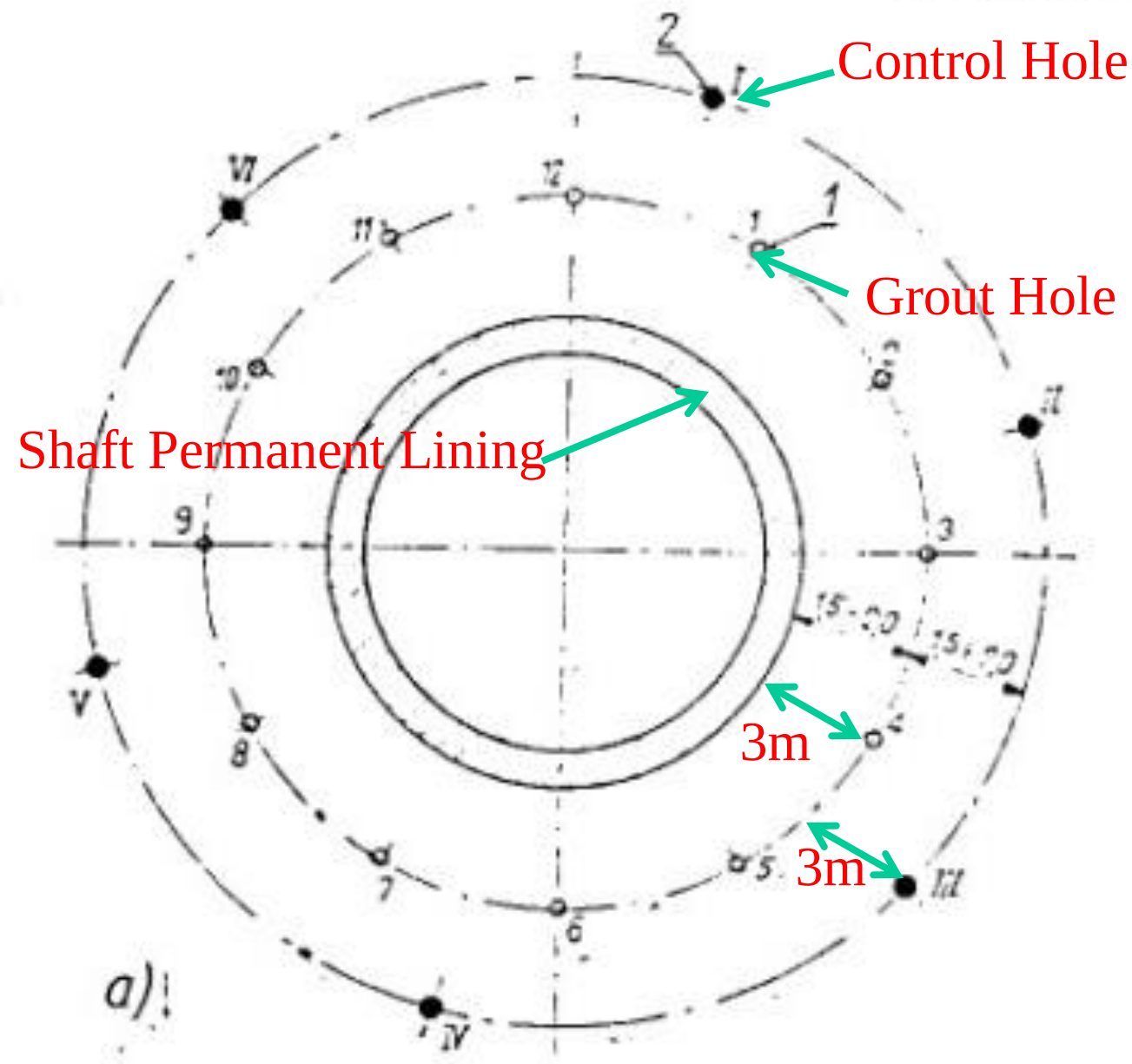
Shaft Sinking Grout Control I

- * Effectiveness of process is assured by measuring water inflow to shaft after grouting & by examining rock core obtained from control diamond drill holes.
- * Grouting operation is successful when control holes prove that water absorbability with in ground rock is less than 0.05L/min/m.
- * Control holes, where grouting operation is carried from surface, have diameter 80-100 mm.
- * Holes are located on opposite sides between grouting holes, in a circle with diameter 3 to 4m longer than circle of grouting holes.

Shaft Sinking Grout Control II

*When grouting from bottom of shaft, control holes are drilled in a circle with in grout holes circle at a diameter distance of 0.6 to 0.8m.

Grout Control Holes From Surface I



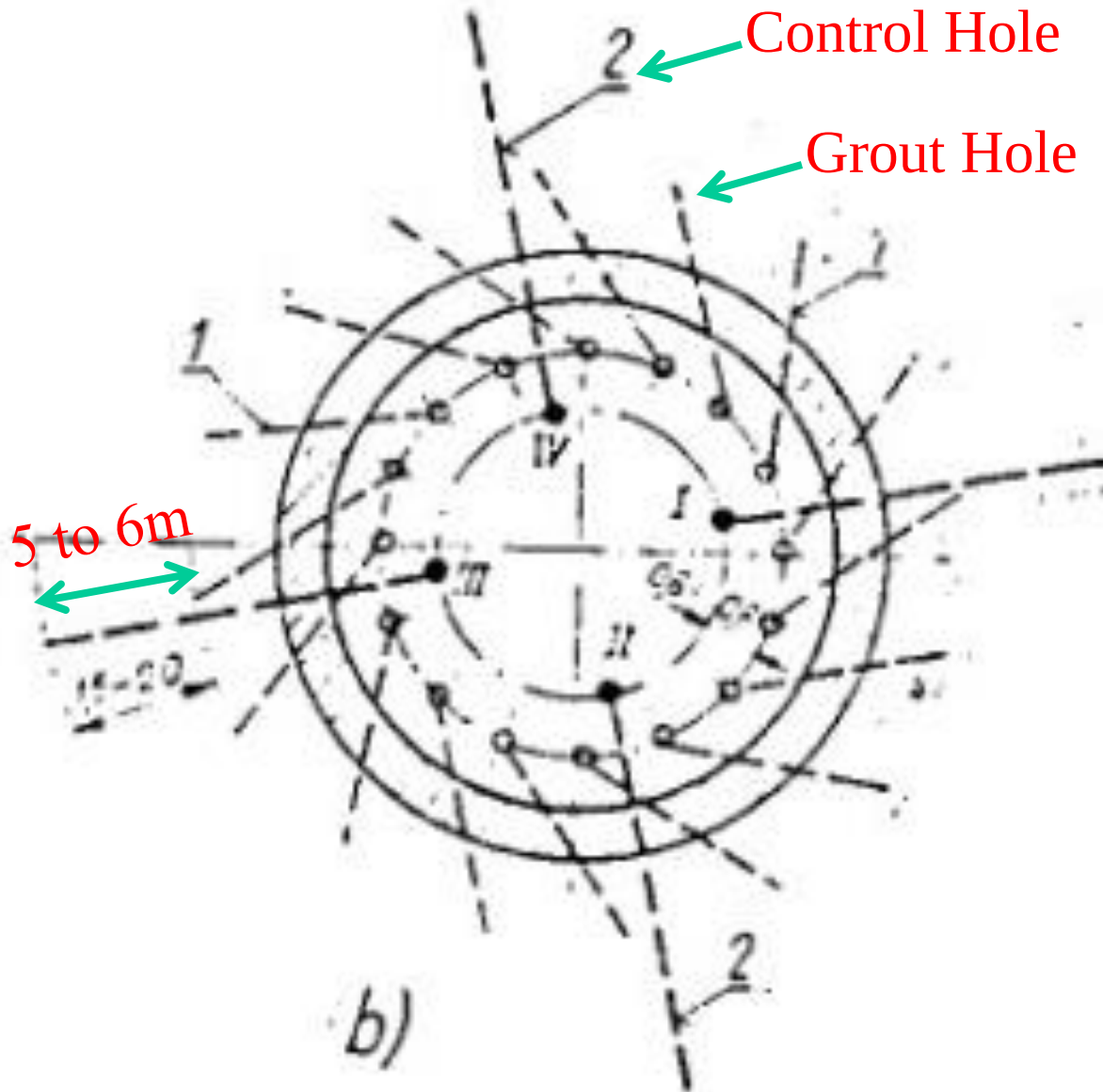
Location of control holes for grouting operation from surface.

1- Grouting hole

2- Control holes

Depth of control holes is same as depth of grout holes.

Grout Control Holes From Shaft



*Location of control holes for grouting from shaft bottom.

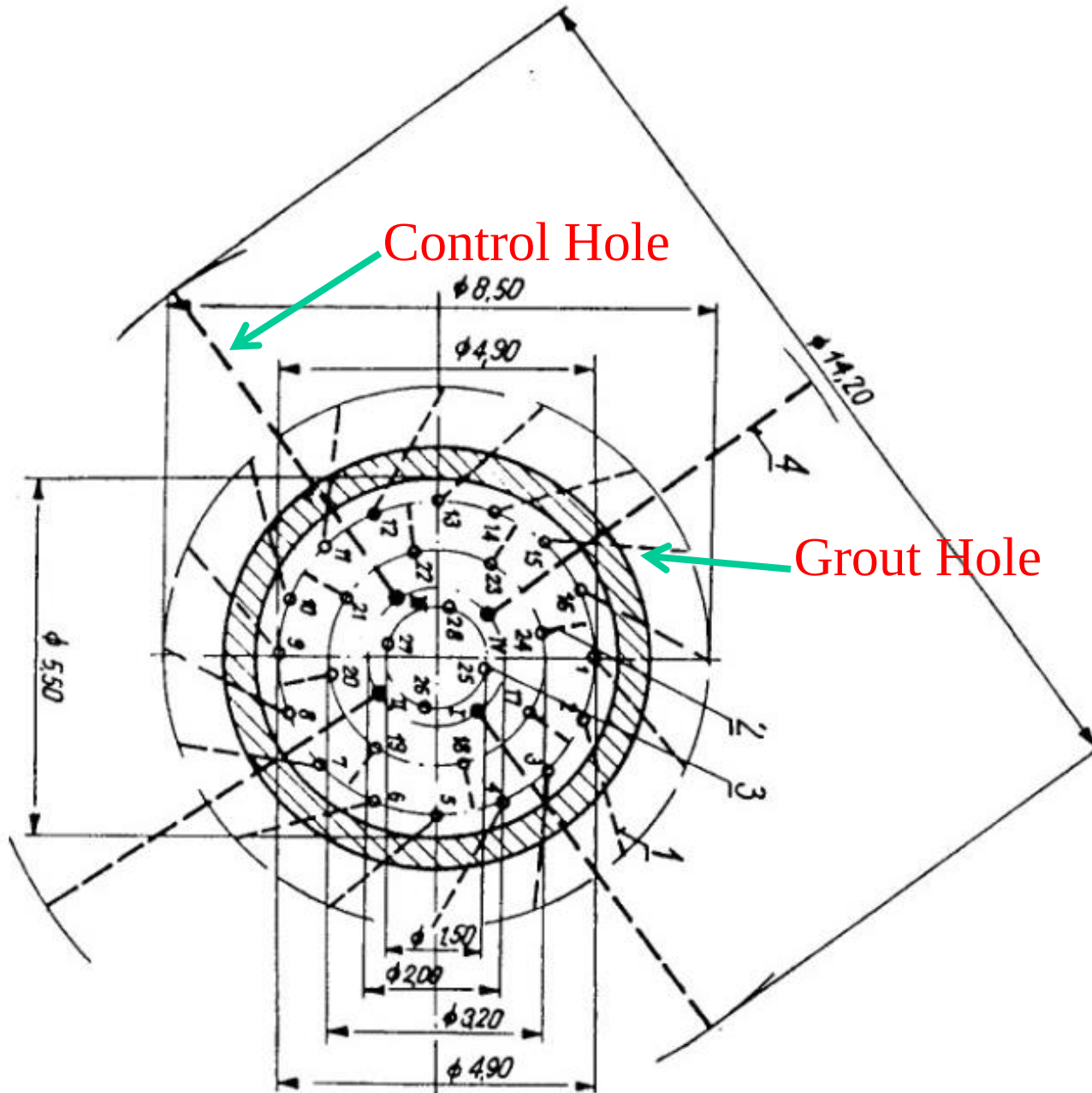
1- Grouting holes

2- Control holes.

Depth of control holes is same as depth of grout holes.

Distribution Grouting & Control Holes

Distribution of grouting & control holes from shaft in 5.5m dia shaft.



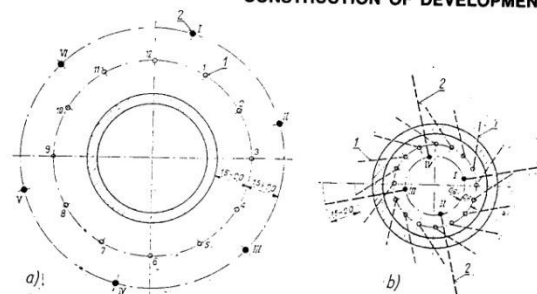


Fig. 17.4.47. Location of control holes.
a) for grouting operation from the surface
b) for grouting from the shaft bottom
1. grouting holes
2. control holes

The depth of the control holes is the same as the depth of the grouting holes. Dimensions in m. Conversion factor: 1 ft = 0.3048 m.

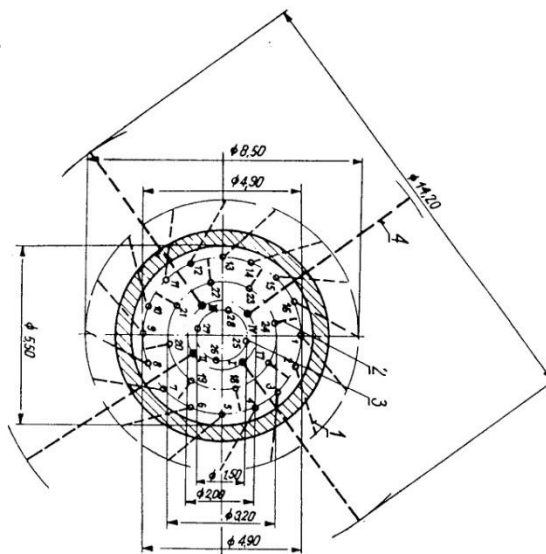


Fig. 17.4.48. Schematic of distribution of grouting and control holes in 18-ft (5.5-m) diameter shaft. Dimensions in m. Conversion factor: 1 ft = 0.3048 m.

The grouting operation is successful when control holes prove that water absorptivity within the grouted rocks is less than 0.04 gpm/ft (0.05 L/min/m).

Control holes, where the grouting operation is carried out from the surface, have diameters of 3 to 4 in. (80 to 100 mm) and are located on opposite sides, between grouting holes, in a circle with a diameter of 9 to 12 ft (3 to 4 m) larger than the circle of grouting holes (Fig. 17.4.47a).

When grouting from the bottom of the shaft, control holes are drilled in a circle within the grout holes circle at a distance of 2 to 2.6 ft (0.6 to 0.8 m) (Fig. 17.4.47b).

Shown in Fig. 17.4.48 is an example of the hole plan for grouting carried out from the shaft bottom with a 36-ft (11-m) height for grouting sections.

GROUT CONSUMPTION. Grout consumption per 1 ft (0.3 m) of shaft varies widely and depends on the fissility of the rocks and the grouting system. The approximate mean figures for grout consumption for designing purposes that can be used are as follows: for sandstone, from 1.1 to 5.5 tons/ft (3.3 to 16.5 t/m); for dolomite, from 0.7 to 1.5 tons/ft (2.2 to 4.4 t/m); for chalk, about 1.3 tons/ft (4.0 t/m); and for diabase from 0.2 to 0.3 tons/ft (0.6 to 1.0 t/m).

EXCAVATION OF CEMENTED ZONE. The excavation method in the cemented zone is similar to the conventional method of shaft sinking. However, the depth of the blasting holes should be limited to 3 to 5 ft (1 to 1.5 m). In addition, the size of the charge in a particular hole and the overall charge being blasted should be reduced.

Shaft Sinking Grout Consumption I

- *Grout consumption per 0.3m of shaft varies widely & depends on feasibility of rock & grouting system.
- *Approximate mean figure for grout consumption for designing purpose that can be used are as follows.
- *Sand Stone – 3.3 to 16.5t/m
- *Dolomite – 2.2 to 4.4t/m
- *Chalk – 4t/m
- *Diabase / dolerite – 0.6-1t/m

Shaft Sinking Grout Zone Excavation I

- *Excavation method in cemented zone is similar to conventional method of shaft sinking.
- *Depth of blasting hole should be limited to 1 to 1.5m.
- *Size of charge in a particular hole & over all charges being blasted should be reduced.
- *Typical explosive consumption on different formation are provided below.
- *Sand Stone – 1.3kg/m^3
- *Shale – 1.1kg/m^3
- *Coal – 0.6kg/m^3

Shaft Sinking Grout Zone Excavation II

- *Excavation proceeds only to a certain depth in a grouted zone.
- *when approaching limit of zone, out going holes to be drilled through valve to test for existence of water hazard.

Shaft Sinking Lining I

- *There is always a possibility of some water penetration in shaft.
- *permanent lining for grout zone should be water tight.

Shaft Sinking Performance & Cost I

- *Sinking with grouting through water bearing strata takes 35 to 50% extra time in compared to normal sinking.
- *Mean monthly advance is in a range of 8-15m/month.
- *Cost of sinking through water bearing strata is about 30% to 50% more than cost of sinking in normal condition.

Shaft Complementary Grouting I

- *Complementary grouting of a shaft lining is to fill up voids between lining & rock mass.
- *Some time complementary grouting is to fill up void in rocks behind shaft lining.
- *It is always done after sinking in difficult hydrological condition.
- *When freezing method is used shaft sinking, complimentary grouting is must.
- *With thawing consolidation of rock disappears.
- *There is lot of change in stress pattern due to volumetric change that cause new cracks in lining.

Shaft Complementary Grouting II

- *Complimentary grouting is beneficial after sinking with normal method.
- *This helps on tightening of lining, filling voids or strengthening rocks.
- *This enable shaft for better service condition.
- *Complimentary grouting also helps shaft to improve condition of existing shaft , where due to multiple reason (corrosion, vibration from shaft vessels, subsidence etc) lining tightness has deteriorated.

Shaft Complementary Grouting III

- *When a large section of shaft is to be grouted, shaft is divided in smaller sections of 15 to 25m long.
- *Sections are grouted downwards, but each section is grouted from bottom to top.
- *In every section on circle holes are drilled 1 to 2 m apart & in vertical direction.
- *Grouting holes are drilled of 30 to 50mm diameter.
- *Depth of holes depends on need, varies from 0.7 to 2m.
- *Holes are located in checkerboard pattern
- *Holes in circle are 1 to 2 m apart.

Shaft Complementary Grouting IV

- *Grouting pressure should exceed hydrostatic water head but should not exceed critical pressure of lining.
- *Assuming short term loading, safety factor for lining can be lowered to 2 or 1.5.
- *In case, first grouting is unsuccessful, process is to be repeated.
- *Tubbing segment have special holes with plug for grouting purposes.
- *Grouting in shaft is usually conducted from hanging platform.

Shaft Complementary Grouting V

*Grouting operation is satisfactory, when water inflow in shaft dies not exceed 25lit/min.

Reference I

*SME Mining Engineering Hand Book, 2nd Edition,
Volume 2, Page 1607 to 1616